In this journal entry, I will share my thoughts and experiences on Module 7 and how I learned about the confusion matrix, precision, and recall. This lab taught me how to calculate a confusion matrix using the Titanic dataset and showed me the 4 quadrants and what they mean. This then led to precision vs. recall in the lab and when you would want to use each of them. The final thing this lab taught me was cross-validation, where each part of the data is used to test and train the machine, not just a random 20%.

The first thing I was shown in the lab was the confusion matrix and how to calculate it. I noticed 4 quadrants in the confusion matrix: 1. True positive, 2. False positive. 3. True Negative 4. False negative. These 4 quadrants show the recall and precision of the model perfectly. If we want a high amount of precision, we would want to look at the true positives and false positives box, as if these are a high percentage, this means the model is correct most of the time, and we would want this for things like a self driving car as if the car sees a crack in the road and believes that this a road line and auto corrects the steering by turning it could hit another car by going into a different lane. This is an example of when someone would want more precision; this is where a computer would flag something if it only had a high chance of being true, as it would waste resources and maybe have consequences if the prediction is wrong. This is why these precision algorithms have a very high cutoff rate of usually 90% or higher to flag something as suspicious. In Module 7, the code that was given was 0.78, which meant that 1 in 5 people in my program who said they survived did not; this is a false positive. This difference showed me that you need to pick a side of the line; if you want one, you can't fully get the other, and the more I wonder how people decide the trade-off they want to get more precision or recall in the model.

On the other hand, recall is the exact opposite in being very sensitive to anything that has a low percentage of actually happening. Think about going to the airport and scanning your bags on the conveyor belt. If an AI system is attached to help identify any dangerous objects, we would want this system to be ultra sensitive, so the threshold is quite low just to be cautious for any danger for most systems. This one we care less about the true positive and true negative quadrants, as even if the chance is small, we don’t want to take a risk. In Module 7, recall was used for detecting survivors and only got a rate of 0.73, which, if we used it in the airport scanner, is not a good rate to detect dangerous objects. This showed me that accuracy alone is misleading, as it showed 0.80 when it was actually 0.73.

The final part of this module was the cross validation method used to test the model. When I first looked at this method in the PowerPoint presentation, I wondered why the standard train-test split was risky, since we have been using it to train our models over the last few modules. I then realized that every time we use the train test split it only tests on 20%. For previous modules, this method was fine, but it is not ideal, as cross-validation trains on all the data and tests on all the data. This happens because the model is tested 5 times; each time, a different 20% of the dataset is tested, and the previous test 20% becomes the train. It's like taking a unit test vs the Final; you only have to study that portion of the entire course, but for the final you have to study every portion of the course.

Module 7 has shown me that evaluating a model takes more than just testing it and seeing the results; it takes multiple factors like Precision, recall, and cross validation to be able to get the result that you want these 3 need to be carefully picked if you want your model to be as efficient as possible.